



Optical Lens

I am looking for Lens Production process with the following specification. 1. Lens diameter : 50~200 mm 2. Lens type: spherical / aspheric/ freeform/ plano 3. material type: optical glass (BK7, fused silica, crown/flint glass, other) 4. surface quality: $R_a \leq 2\text{nm}$ 5. Automation: full or semi 6. Coating: AR/Hard, multi-layer

I'll map this as a production workflow first, then list the equipment/process options that fit 50–200 mm optical glass lenses with $R_a \leq 2\text{ nm}$ and AR/hard multilayer coating.

Thought for 13s

Below is a practical **lens production process** for your specification:

Target specification

Item	Requirement
Lens diameter	50–200 mm
Lens types	Spherical, aspheric, freeform, plano
Materials	BK7, fused silica, crown/flint glass, other optical glass
Surface roughness	$R_a \leq 2\text{ nm}$
Automation	Semi-automatic or full automatic
Coating	AR / hard coating / multilayer coating

Recommended Production Flow

1. Raw Glass Selection

Use optical-grade glass blanks from suppliers such as Schott, Ohara, Hoya, CDGM, Corning, or equivalent.

Typical materials:

Material	Use case
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BK7 / N-BK7	General optical lenses, imaging optics
Fused silica	UV, laser, high thermal stability
Crown glass	Low dispersion, general optics
Flint glass	High refractive index, achromatic systems
Other optical glass	Custom refractive index / Abbe number requirement

Incoming inspection should check **diameter, thickness, bubbles, striae, refractive index, internal stress, and material certificate.**

2. Cutting / Blanking

Large glass blocks or rods are cut into lens blanks.

Equipment

Process	Machine
Slicing	Diamond wire saw / inner-diameter saw
Rough blanking	CNC glass cutting machine
Edge preforming	CNC edging machine

For 50–200 mm lenses, use diamond tooling and coolant to avoid chipping and thermal damage.

3. Curve Generation / Rough Grinding

This step creates the basic spherical, aspheric, freeform, or plano geometry.

Lens type	Recommended process
Plano	Double-side grinding / lapping
Spherical	CNC curve generator
Aspheric	CNC grinding / ultra-precision grinding
Freeform	5-axis CNC grinding / deterministic grinding

For aspheric and freeform lenses, CNC grinding is preferred because conventional spherical generators cannot produce complex non-rotational forms.

A common modern optics workflow is **milling/grinding → polishing → deterministic correction such as MRF or ion-beam figuring**. MRF is widely used for localized correction of spheres, aspheres, and freeform optics, while ion-beam polishing/figuring is used when very high form accuracy and low roughness are required.

4. Fine Grinding

Fine grinding removes subsurface damage from rough grinding and prepares the surface for polishing.

Typical abrasive sequence:

Stage	Abrasive size
Rough grinding	80–180 grit diamond
Medium grinding	320–600 grit diamond
Fine grinding	9–25 µm diamond
Pre-polish	3–9 µm diamond / alumina

Goal before polishing:

Parameter	Target
Form error	Within polishing allowance
Subsurface damage	Minimized
Surface condition	No deep scratches / chips
Edge condition	Safe for polishing and coating

5. Polishing

This is the key stage for reaching **Ra ≤ 2 nm**.

Recommended polishing methods

Lens type	Suitable polishing method
Plano	Double-side polishing / pitch polishing / CMP
Spherical	CNC polishing / pitch polishing
Aspheric	CNC sub-aperture polishing / MRF
Freeform	5-axis sub-aperture polishing / MRF / ion beam figuring

For your requirement, I would not rely only on basic polishing. Use this sequence:

1	Fine grinding
2	→ CNC pre-polishing
3	→ deterministic polishing
4	→ MRF / ion beam correction
5	→ final cleaning

MRF can correct local form errors with nanometer-level control and is suitable for complex shapes including spheres, aspheres, and freeforms. Some commercial precision optics polishing systems support optics in the 5–200 mm range, which matches your requested 50–200 mm diameter range.

For very high-end optics, **MRF and ion-beam polishing/figuring** are commonly used as final refinement processes. Reported achievable roughness can be below 1–2 nm depending on material, process control, and inspection method.

6. Centering and Edging

After polishing, the lens outside diameter and optical axis are aligned.

Operation	Purpose
Centering	Align mechanical diameter to optical axis
Edging	Final OD control
Chamfering / beveling	Prevent edge chips
Blackening, if needed	Reduce stray light

For 50–200 mm lenses, CNC centering machines are preferred.

7. Cleaning Before Coating

This is critical. Coating defects often come from contamination, not coating design.

Recommended cleaning flow:

- 1 Ultrasonic cleaning
- 2 → alkaline wash
- 3 → DI water rinse
- 4 → solvent rinse if required
- 5 → filtered drying
- 6 → plasma / ion cleaning before coating

Use a cleanroom or clean coating preparation area. For high-quality multilayer AR, Class 1000 or better is often preferred around coating preparation.

8. Coating Process

Your coating requirement is:

- 1 AR coating
- 2 Hard coating
- 3 Multilayer coating

For optical glass lenses, the best production method is usually **vacuum thin-film coating**.

Recommended coating technologies

Coating type	Recommended method
Single-layer AR	Vacuum evaporation / sputtering
Broadband multilayer AR	Ion-assisted deposition / magnetron sputtering
Hard protective layer	SiO ₂ -based hard layer / DLC-like layer depending on use
High durability coating	Ion-assisted deposition or sputtering
Laser optics coating	IBS / IAD / advanced sputtering

Optical coatings are thin films used to modify reflection, transmission, or absorption, and multilayer AR coatings are commonly designed using alternating high- and low-index materials such as TiO₂, Ta₂O₅, Nb₂O₅, Al₂O₃, MgF₂, and SiO₂.

Recommended coating stack example:

- 1 | Glass substrate
- 2 | → adhesion layer
- 3 | → hard layer, if required
- 4 | → multilayer AR stack
- 5 | → hydrophobic / oleophobic top coat, optional

For glass lenses, “hard coating” may mean different things depending on the industry. Plastic lenses usually need a hard coat for scratch resistance. Glass already has higher hardness, so for glass optics the hard/durable coating is usually a protective dielectric layer or durable AR stack rather than a thick polymer hard coat.

9. Final Inspection

For your target, inspection must include both geometry and surface quality.

Required inspection items

Inspection item	Recommended equipment
Diameter / thickness	Digital gauge / CMM
Radius of curvature	Radius bench / interferometer
Surface form	Fizeau interferometer / CGH / stitching interferometer
Aspheric/freeform form	Profilometer / CMM / interferometric stitching
Surface roughness Ra	White-light interferometer / AFM
Scratch-dig	Visual inspection / automated defect inspection
Transmittance	Spectrophotometer
Coating reflection	Spectrophotometer
Coating adhesion	Tape test / environmental test
Coating durability	Abrasion / humidity / temperature cycling

For drawing and quality specification, use **ISO 10110** for optical drawings and surface texture. ISO 10110-8 is commonly used for specifying surface roughness and waviness on polished optics.

For scratch/dig visual surface quality, many optics suppliers still use **MIL-PRF-13830B**, commonly written as 60-40, 40-20, 20-10, etc. Scratch-dig is different from Ra: Ra measures roughness; scratch-dig measures visible surface defects.

Recommended Production Route by Lens Type

A. Plano Lens

1	Glass blank
2	→ slicing
3	→ double-side grinding
4	→ double-side polishing
5	→ edge grinding
6	→ cleaning
7	→ AR / hard multilayer coating
8	→ inspection

Best for windows, filters, flat lenses, protective optics.

B. Spherical Lens

1	Glass blank
2	→ curve generation
3	→ fine grinding
4	→ spherical polishing
5	→ centering / edging
6	→ cleaning
7	→ AR / hard multilayer coating
8	→ inspection

Best for standard convex/concave lenses.

C. Aspheric Lens

1	Glass blank
2	→ CNC aspheric grinding
3	→ fine grinding
4	→ CNC sub-aperture polishing
5	→ MRF correction
6	→ centering / edging
7	→ cleaning
8	→ AR / hard multilayer coating
9	→ interferometric inspection

Best for high-performance imaging optics, laser optics, projection systems.

D. Freeform Lens

1	Glass blank
2	→ 5-axis CNC grinding
3	→ sub-aperture polishing

```
4 → MRF / ion beam figuring
5 → metrology feedback loop
6 → centering / edging
7 → cleaning
8 → AR / hard multilayer coating
9 → freeform inspection
```

Freeform lenses need a **closed-loop metrology process**. That means:

```
1 | Polish
2 | → measure
3 | → calculate error map
4 | → correct surface
5 | → measure again
```

This loop may repeat several times.

Automation Recommendation

For your specification, I recommend **semi-automatic to full automatic production**, depending on volume.

Semi-Automatic Line

Best for low/mid volume, high mix, custom optics.

```
1 | Operator loads glass blank
2 | → CNC grinding
3 | → CNC polishing
4 | → MRF correction
5 | → manual/automatic inspection
6 | → batch coating
```

Advantages:

Advantage	Explanation
Flexible	Good for spherical, aspheric, freeform, plano
Lower investment	Cheaper than full automation
Good for custom optics	Easier process changes

Disadvantage: more operator skill required.

Full-Automatic Line

Best for mass production.

```
1 | Robot loading
```


- 2 → CNC grinding
- 3 → automated washing
- 4 → automated polishing
- 5 → automated metrology
- 6 → automatic correction loop
- 7 → robotic coating loading
- 8 → automatic inspection

Advantages:

Advantage	Explanation
Stable quality	Less operator variation
Higher throughput	Better for volume production
Traceability	Easier process data collection

Disadvantage: expensive and less flexible for many custom shapes.

Recommended Equipment List

Process	Equipment
Cutting	Diamond wire saw / ID saw
Curve generation	CNC lens generator
Fine grinding	CNC fine grinder
Plano processing	Double-side lapping / polishing machine
Spherical polishing	CNC spherical polisher
Aspheric/freeform polishing	CNC sub-aperture polisher
Nanometer correction	MRF machine / ion beam figuring
Centering	CNC centering and edging machine
Cleaning	Ultrasonic cleaning line
Coating	Vacuum AR coating machine / IAD / IBS / sputtering system
Roughness inspection	White-light interferometer / AFM
Form inspection	Fizeau interferometer / CGH / profilometer

Coating inspection	Spectrophotometer
Defect inspection	Scratch-dig inspection system

Key Technical Notes

For **Ra ≤ 2 nm**, the most important stages are:

- 1 fine grinding quality
- 2 + polishing slurry selection
- 3 + polishing pad/tool control
- 4 + temperature stability
- 5 + cleanroom handling
- 6 + final MRF/IBF correction
- 7 + accurate roughness measurement

For **aspheric and freeform lenses**, the most important factor is not only roughness. You also need to control:

- 1 surface form error
- 2 mid-spatial frequency error
- 3 slope error
- 4 tool marks
- 5 subsurface damage
- 6 coating stress

For **coating**, define exact optical requirements before production:

- 1 wavelength range
- 2 average reflectance
- 3 maximum reflectance
- 4 angle of incidence
- 5 polarization
- 6 laser damage threshold, if laser use
- 7 environmental durability
- 8 adhesion requirement
- 9 abrasion resistance

Suggested Final Process Specification

You can write your process requirement like this:

- 1 Production process shall support optical glass lens manufacturing for spherical,
- 2 aspheric, freeform, and plano lenses with diameter range 50-200 mm.
- 3
- 4 Materials shall include BK7, fused silica, crown glass, flint glass, and equivalent optical glass materials.
- 5
- 6
- 7 The process shall include glass blank cutting, CNC curve generation, fine grinding,
- 8 CNC polishing, deterministic correction polishing such as MRF or ion beam figuring

```
9 where required, centering/edging, ultrasonic cleaning, vacuum
multilayer AR/hard
10 coating, and final optical inspection.
11
12 Final polished surface roughness shall be  $R_a \leq 2$  nm before coating
unless otherwise
13 specified. Surface roughness shall be measured using white-light
interferometry,
14 AFM, or equivalent calibrated metrology.
15
16 The production line shall support semi-automatic or full-automatic
operation,
17 including CNC process control, metrology feedback, batch traceability,
and coating
18 process recipe control.
19
20 Coating shall support AR, hard protective coating, and multilayer
dielectric coating
21 according to customer wavelength, angle, durability, and environmental
requirements.
```

My recommendation: for your spec, choose **CNC grinding + CNC polishing + MRF correction + vacuum multilayer coating** as the baseline process. For freeform or very high-end aspheric lenses, add **ion beam figuring** and closed-loop metrology.

good for the recommendation. please give me more detailed info for the CNC grinding + CNC polishing + MRF correction + vacuum multilayer coating

Good, let's zoom into the actual production chain: what each stage does, what machines/tools are involved, what parameters matter, and what acceptance checks you should define for vendors.

Thought for 4s

Yes — this is the right process chain for **high-precision optical glass lenses**:

```
1 CNC grinding
2 → CNC polishing
3 → MRF correction
4 → vacuum multilayer coating
```

For your target, this chain is suitable for:

```
1 Diameter: 50-200 mm
2 Lens type: spherical / aspheric / freeform / plano
3 Material: BK7, fused silica, crown/flint optical glass
4 Surface roughness:  $R_a \leq 2$  nm
5 Automation: semi or full automation
6 Coating: AR / hard / multilayer
```

1. CNC Grinding

Purpose

CNC grinding creates the basic optical shape before polishing.

It controls:

- 1 radius
- 2 thickness
- 3 sag
- 4 wedge
- 5 aspheric profile
- 6 freeform profile
- 7 surface allowance for polishing

This is not the final surface-quality step. CNC grinding gives the lens the correct geometry, but the surface still has grinding marks and subsurface damage.

Suitable Lens Types

Lens type	CNC grinding method
Plano	CNC flat grinding / double-side grinding
Spherical	CNC curve generation
Aspheric	CNC aspheric grinding
Freeform	5-axis CNC grinding
Meniscus	CNC curve generation both sides

For **spherical lenses**, the machine follows a rotationally symmetric curve.

For **aspheric lenses**, the machine follows a programmed sag equation.

For **freeform lenses**, the machine needs multi-axis tool-path control because the surface is not simply rotationally symmetric.

Typical CNC Grinding Flow

- 1 Glass blank loading
- 2 → rough centering
- 3 → rough curve generation
- 4 → semi-fine grinding
- 5 → fine grinding
- 6 → washing
- 7 → dimensional inspection

Key Process Parameters

Parameter	Why it matters
Spindle speed	Affects removal rate and surface damage
Feed rate	Too high causes cracks/chipping
Diamond wheel grit	Controls roughness and subsurface damage
Coolant flow	Prevents thermal stress
Tool path	Critical for aspheric/freeform accuracy
Grinding allowance	Must leave enough material for polishing
Wheel dressing condition	Affects shape accuracy and scratch risk

Recommended Grinding Stages

Stage	Tool	Purpose
Rough grinding	Coarse diamond wheel	Fast material removal
Semi-fine grinding	Medium diamond wheel	Shape refinement
Fine grinding	Fine diamond wheel	Reduce subsurface damage
Pre-polish prep	Very fine diamond / lapping	Prepare for polishing

A good sequence is:

```
1 | coarse diamond
2 | → medium diamond
3 | → fine diamond
4 | → very fine diamond
5 | → polishing
```

Do not jump from rough grinding directly to polishing. That causes long polishing time and may leave hidden subsurface damage.

Output After CNC Grinding

Typical target before polishing:

Item	Recommended target
Surface condition	Uniform fine-ground surface
Chips	No edge breakout beyond tolerance
Form error	Within polishing correction range
Subsurface damage	Minimized
Thickness	Near final thickness with polishing allowance
Roughness	Not final; polishing will improve it

For your final **Ra ≤ 2 nm**, grinding must be stable. If grinding damage is too deep, polishing may look shiny but still have micro-defects or stress damage.

2. CNC Polishing

Purpose

CNC polishing removes fine-grinding marks and brings the lens close to final roughness and form.

It improves:

- 1 surface roughness
- 2 transparency
- 3 form accuracy
- 4 slope error
- 5 mid-spatial frequency error

CNC polishing is more controlled than manual pitch polishing, especially for aspheric and freeform optics.

Suitable Polishing Methods

Lens type	Recommended polishing
Plano	Double-side polishing / CNC flat polishing
Spherical	CNC spherical polishing
Aspheric	CNC sub-aperture polishing

Freeform	5-axis CNC sub-aperture polishing
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For **plano lenses**, double-side polishing is often efficient.

For **spherical lenses**, conventional CNC polishing or pitch polishing can work well.

For **aspheric/freeform lenses**, use **computer-controlled sub-aperture polishing**.

Typical CNC Polishing Flow

1	Load fine-ground lens
2	→ polish first side
3	→ clean
4	→ measure
5	→ polish correction pass
6	→ polish second side
7	→ clean
8	→ measure again

For aspheric/freeform optics:

1	Polish
2	→ measure surface error map
3	→ generate correction tool path
4	→ polish again
5	→ measure again

This is a closed-loop process.

Key Process Parameters

Parameter	Importance
Polishing pad/tool material	Affects removal stability
Slurry type	Affects roughness and removal rate
Slurry particle size	Too large increases roughness
Tool pressure	Too high causes deformation or edge roll-off
Dwell time	Controls local material removal
Tool path	Critical for aspheric/freeform correction
Temperature	Thermal drift affects precision
Cleaning between steps	Prevents scratches

Common Polishing Slurries

Material	Common polishing slurry
BK7 / crown glass	Cerium oxide
Fused silica	Cerium oxide or colloidal silica
Flint glass	Cerium oxide, carefully controlled
Hard glass	Cerium oxide / diamond pre-polish

For **Ra ≤ 2 nm**, the final slurry and pad condition are very important. Dirty slurry or worn pads can create micro-scratches.

Output After CNC Polishing

Recommended target before MRF:

Item	Target
Surface roughness	Near or below Ra 2 nm if possible
Form error	Close enough for MRF correction
Visible defects	No scratches, digs, streaks
Edge condition	No chips
Mid-spatial error	Controlled

CNC polishing should not be used as the only final correction step for demanding aspheric/freeform lenses. MRF is better for deterministic final correction.

3. MRF Correction

What MRF Means

MRF = Magnetorheological Finishing.

It is a deterministic polishing/correction process using a magnetically controlled polishing fluid. The fluid becomes a controlled polishing “spot” under a magnetic field and removes material

very precisely.

MRF is used when conventional polishing is not accurate enough, especially for **aspheric, freeform, high-precision spherical, and plano optics**. It is considered a deterministic surface finishing process because the machine can calculate where and how long to polish based on the measured error map.

Why Use MRF After CNC Polishing?

CNC polishing improves the surface, but it can leave:

```
1 | small form error
2 | local high/low zones
3 | mid-spatial error
4 | edge roll-off
5 | tool marks
6 | slope error
```

MRF corrects these errors locally.

The basic logic is:

```
1 | Measure lens surface
2 | → create error map
3 | → calculate dwell-time map
4 | → MRF removes high spots
5 | → measure again
6 | → repeat if needed
```

This is why MRF is excellent for high-end optics.

MRF Process Flow

```
1 | Clean polished lens
2 | → mount lens in MRF machine
3 | → measure initial surface error
4 | → import measurement/error map
5 | → generate dwell-time correction program
6 | → run MRF correction
7 | → clean lens
8 | → inspect surface again
9 | → repeat correction if required
```

For freeform optics, the measurement and correction loop is especially important. Advanced freeform production commonly combines CNC machining, MRF polishing, and 3D metrology verification.

MRF Strengths

Strength	Explanation
Very precise local correction	Good for removing small form errors
Good for aspheres	Can correct non-spherical profiles
Good for freeforms	Works with measured error map
Low subsurface damage	Flexible polishing spot
Repeatable	Computer-controlled dwell time
Supports high-end optics	Useful for precision imaging and laser optics

MRF can achieve very high precision. Some optics manufacturers describe MRF as capable of producing figure error down to nanometer-level RMS and very low roughness, depending on material, geometry, and process control.

MRF Limitations

Limitation	Explanation
Not for heavy material removal	CNC grinding/polishing must do bulk removal first
Requires accurate metrology	Bad measurement gives bad correction
Slower than conventional polishing	Best for precision correction
Edge zones can be difficult	Edge correction needs careful setup
Machine cost is high	Usually used for high-value optics

MRF should be treated as a **final correction process**, not a replacement for grinding and polishing.

Best Use of MRF in Your Case

For your lens range **50–200 mm**, MRF is highly suitable when:

- 1 surface form tolerance is tight
- 2 aspheric/freeform geometry is required
- 3 $R_a \leq 2 \text{ nm}$ is required
- 4 mid-spatial error must be controlled
- 5 coating quality must be high

Recommended use:

Lens type	MRF needed?
Plano standard lens	Optional
High-precision plano	Recommended
Spherical standard lens	Optional
High-precision spherical	Recommended
Aspheric lens	Strongly recommended
Freeform lens	Very strongly recommended

4. Vacuum Multilayer Coating

Purpose

After polishing and MRF correction, the lens is coated to improve optical performance and durability.

Your requested coating types:

1

AR coating

2

hard / durable coating

3

multilayer coating

For optical glass, this is normally done by **vacuum thin-film coating**.

Why Vacuum Coating?

Vacuum coating gives controlled thin-film layers with precise optical thickness.

A multilayer AR coating works by using interference between reflections from different layers to reduce total reflection. Commercial optical coatings are commonly produced by methods such as thermal evaporation, ion-assisted deposition, and ion beam sputtering; the method affects durability, stress, coating stability, and laser damage performance.

Recommended Coating Methods

Method	Best use
--------	----------

Thermal evaporation	Lower-cost standard AR
E-beam evaporation	General precision optical coating
Ion-assisted deposition, IAD	Durable AR, better density
Ion beam sputtering, IBS	High-end laser / precision coatings
Magnetron sputtering	Durable industrial optical coating

For your requirement, I recommend:

- 1 IAD coating for general high-quality AR/hard multilayer coating
- 2 IBS coating for highest performance laser or ultra-stable optical coating

Ion-assisted e-beam evaporation uses an electron beam to vaporize coating material in vacuum; the vapor condenses on the optical surface to form controlled layers. Edmund Optics notes that IAD e-beam coatings can provide uniform, low-stress layers and are used for UV and NIR laser-related optics.

Typical Vacuum Coating Flow

- 1 Final polishing inspection
- 2 → ultrasonic cleaning
- 3 → DI water rinse
- 4 → drying
- 5 → visual inspection
- 6 → load into coating fixture
- 7 → vacuum pump-down
- 8 → plasma/ion cleaning
- 9 → deposit multilayer stack
- 10 → optical monitoring / quartz monitoring
- 11 → cooling
- 12 → unload
- 13 → final inspection

Common Multilayer AR Stack Structure

A simplified example:

- 1 Optical glass substrate
- 2 → adhesion / matching layer
- 3 → high-index layer
- 4 → low-index layer
- 5 → high-index layer
- 6 → low-index layer
- 7 → protective top layer

Typical coating materials:

Layer type	Common materials
Low-index layer	SiO ₂ , MgF ₂
High-index layer	TiO ₂ , Ta ₂ O ₅ , Nb ₂ O ₅ , HfO ₂
Hard/protective layer	SiO ₂ , Al ₂ O ₃ , DLC-like layer depending on application
Hydrophobic top layer	Fluorinated top coat, optional

For high-durability AR, use dense oxide layers and ion-assisted deposition.

Coating Design Inputs You Must Define

Before coating, do not just say “AR coating.” You need these details:

Coating requirement	Example
Wavelength range	400–700 nm, 650–1050 nm, 1064 nm, etc.
Reflectance target	Ravg < 0.5% per surface
Angle of incidence	0°, 0–30°, 45°
Polarization	S, P, or unpolarized
Substrate material	BK7, fused silica, etc.
Environment	indoor, outdoor, high humidity, military, medical
Laser use	wavelength, pulse width, power density
Durability	abrasion, humidity, salt fog, temperature cycle
Coating side	one side or both sides

Without these, the coating vendor cannot design the correct stack.

Recommended Full Process Route

For Spherical Lens

- 1 Optical glass blank
- 2 → CNC curve generation

- 3 → fine grinding
- 4 → CNC spherical polishing
- 5 → surface/form inspection
- 6 → MRF correction if high precision required
- 7 → centering/edging
- 8 → ultrasonic cleaning
- 9 → vacuum AR/hard multilayer coating
- 10 → final inspection

For Aspheric Lens

- 1 Optical glass blank
- 2 → CNC aspheric grinding
- 3 → fine grinding
- 4 → CNC sub-aperture polishing
- 5 → interferometric/profile measurement
- 6 → MRF correction
- 7 → repeat measurement/correction loop
- 8 → centering/edging
- 9 → ultrasonic cleaning
- 10 → vacuum multilayer AR/hard coating
- 11 → final inspection

For Freeform Lens

- 1 Optical glass blank
- 2 → 5-axis CNC grinding
- 3 → CNC sub-aperture polishing
- 4 → 3D surface measurement
- 5 → error-map generation
- 6 → MRF correction
- 7 → repeat metrology correction loop
- 8 → centering/edging
- 9 → ultrasonic cleaning
- 10 → vacuum multilayer coating
- 11 → final freeform inspection

For Plano Lens

- 1 Optical glass blank
- 2 → slicing
- 3 → double-side grinding
- 4 → double-side polishing
- 5 → MRF correction if high flatness required
- 6 → edging/chamfering
- 7 → cleaning
- 8 → vacuum multilayer coating
- 9 → final inspection

Suggested Acceptance Criteria

For supplier communication, use this kind of requirement:

Area	Requirement
------	-------------

Diameter	50–200 mm
Material	BK7, fused silica, crown/flint optical glass
Lens geometry	Spherical, aspheric, freeform, plano
Surface roughness	$Ra \leq 2$ nm before coating
Coating	AR / hard / multilayer dielectric
Production process	CNC grinding + CNC polishing + MRF correction + vacuum coating
Automation	Semi-automatic or full automatic
Inspection	Interferometer, profilometer, white-light interferometer/AFM, spectrophotometer
Traceability	Lot number, material certificate, process record, coating recipe ID
Cleanliness	Cleanroom or clean coating preparation area

Recommended Equipment Configuration

Minimum semi-automatic production line

Step	Equipment
Cutting	Diamond saw
Grinding	CNC lens grinder / curve generator
Polishing	CNC polisher
MRF	MRF correction machine
Cleaning	Ultrasonic cleaning line
Coating	Vacuum coating machine with IAD
Inspection	Interferometer + roughness meter + spectrophotometer

This is good for:

- 1

2

3

4

5
- custom optics
small/mid volume
many lens types
engineering production
high-mix production

Full automatic production line

Step	Equipment
Loading	Robot handling
Grinding	CNC grinding cell
Polishing	CNC polishing cell
Inspection	Automated metrology station
Correction	MRF correction cell
Cleaning	Automated ultrasonic cleaning
Coating	Automated vacuum coating system
Data	MES / traceability system

This is good for:

- 1

2

3

4

5
- large volume
stable product family
repeated lens designs
lower labor variation
strong quality traceability

Important Vendor Questions

When you talk to machine or production vendors, ask these:

- 1

2

3

4

5

6

7

8

9

10
1. Can your CNC grinding system process 50-200 mm optical glass lenses?
2. Can it support spherical, aspheric, freeform, and plano geometries?
3. What materials are proven: BK7, fused silica, crown/flint?
4. What roughness can be achieved after CNC polishing?
5. What final roughness can be achieved after MRF?
6. What metrology is integrated with the MRF correction loop?
7. Can the process achieve $Ra \leq 2$ nm repeatedly, not only once?
8. What is the maximum surface form accuracy after MRF?
9. What coating technology is used: evaporation, IAD, IBS, sputtering?
10. Can you provide AR/hard multilayer coating design for our wavelength range?


```
11 11. Can you provide coating durability test reports?
12 12. Can you provide sample inspection reports?
13 13. Is the production semi-automatic or fully automatic?
14 14. What process data is logged for traceability?
15 15. What is the expected yield for 50-200 mm aspheric/freeform glass lenses?
```

My Practical Recommendation

For your spec, I would define the production route like this:

```
1 Primary forming:
2 CNC grinding / 5-axis grinding
3
4 Main finishing:
5 CNC polishing / sub-aperture polishing
6
7 Precision correction:
8 MRF correction with closed-loop metrology
9
10 Coating:
11 Vacuum multilayer AR/hard coating using IAD as standard,
12 IBS for high-end laser or ultra-stable optical requirements.
```

Best default choice:

```
1 CNC grinding + CNC sub-aperture polishing + MRF + IAD vacuum coating
```

Premium choice:

```
1 5-axis CNC grinding + CNC polishing + MRF + IBS coating
```

For **spherical/plano**, the default process is enough.

For **aspheric/freeform**, choose the premium process or at least make sure the vendor has real MRF + freeform metrology capability.

okay, give me the equipment list and recent prices for it

I'll build this as a procurement-style list: equipment category, what specs to ask for, typical suppliers/models, and realistic 2026 budget ranges. For this kind of optics machinery, many prices are quote-only, so I'll separate "public/listed price found" from "industry budget estimate."

Thought for 19s

Below is a **practical equipment list + recent price range** for a production line using:

```
1 CNC grinding → CNC polishing → MRF correction → vacuum multilayer coating
```

Important note: for serious precision-optics equipment, most premium suppliers do **quote-only pricing**. So the prices below are a mix of **public used-machine listings**, **China-market/public marketplace pricing**, and **industry budget ranges**. Treat them as procurement planning numbers, not final quotations.

1. Recommended Core Equipment List

A. CNC Optical Grinding Machine

Used for rough/fine generation of spherical, aspheric, freeform, and plano surfaces.

Item	Recommended spec
Workpiece diameter	50–200 mm minimum
Axis	4-axis or 5-axis
Lens support	Spherical, aspheric, freeform
Material	BK7, fused silica, crown/flint glass
Coolant	Temperature-controlled coolant
Automation	Semi-auto or robotic loading optional

Example suppliers / models

Supplier	Example model	Notes
Satisloh	SPM-200	5-axis CNC grinder for Ø 5–200 mm, spheres/aspheres/freeforms
OptoTech	ASM / SM / GI series	German precision optics machines
Schneider Optical Machines	CNC generators	Strong in ophthalmic/precision optics
Chinese suppliers	Custom CNC optical grinder	Lower price, verify precision carefully

Satisloh's SPM-200 matches your size requirement very well: it is listed as a 5-axis CNC optical grinding machine for Ø 5–200 mm workpieces and supports spheres, aspheres, and freeform surfaces.

Estimated price

Market level	Price range
Used Western/Japanese/Korean precision grinder	\$80,000–250,000
New China-made CNC optical grinder	\$80,000–300,000
New premium Satisloh / OptoTech / Schneider class	\$300,000–800,000+

B. CNC Optical Polishing Machine

Used after grinding to remove grinding marks and improve roughness/form.

Item	Recommended spec
Workpiece diameter	50–200 mm
Polishing type	Sub-aperture polishing preferred
Geometry	Spherical, aspheric, freeform
Software	Dwell-time correction / tool-path control
Metrology link	Error-map import preferred

Example suppliers / models

Supplier	Example model	Notes
Satisloh	SPS-200	5-axis CNC polishing for spheres/aspheres/freeforms
Satisloh	SPS-150	Good for spherical lenses up to Ø 150 mm

OptiPro	PRO 160 UFF	Aspheric polishing up to Ø 160 mm
OptoTech	MCP / ASP series	Spherical/aspheric/freeform polishing
Zeeko	IRP / polishing platforms	High-end freeform polishing

Satisloh lists the SPS-200 as a 5-axis CNC polishing machine for spheres, aspheres, and freeform surfaces; the related SPM/SPS-200 documentation shows a working range that reaches your 50–200 mm target class.

OptiPro's PRO 160 UFF is for precision aspheres up to 160 mm, so it fits part of your requested range but not the full 200 mm.

Estimated price

Market level	Price range
Small/simple China optical polishing machine	\$8,000–20,000 publicly listed for basic ceramic/optical glass polishing machines
Used CNC aspheric/freeform polishing machine	\$10,000–150,000+ depending condition; one used OptiPro UFF-150 listing was shown at \$9,500 , but that is unusually low and should be inspected carefully.
New China-made CNC optical polisher	\$50,000–250,000
New premium CNC polishing machine	\$250,000–700,000+

C. MRF Correction Machine

Used for deterministic nanometer-level correction after CNC polishing.

Item	Recommended spec
Workpiece diameter	At least Ø 200 mm
Modes	Rotational + raster/freeform mode
Geometry	Plano, spherical, aspheric, freeform

Software	Dwell-time correction from measured error map
Tool heads	Multiple spot sizes preferred
Metrology integration	Must support interferometer/profiler error-map import

Example suppliers / models

Supplier	Example model	Notes
QED Technologies	Q-flex MRF-300	Strong fit for Ø up to about 300 mm
QED Technologies	Q-flex 100	Smaller optics, may be limited depending lens size
Raytek / Chinese MRF suppliers	Optical glass MRF machine	Lower-cost option; verify capability
ZR Optics / other China suppliers	MRF equipment	Quote-only, must validate performance

QED's Q-flex MRF-300 is designed for larger optics and production throughput. Its spec sheet states part sizes up to about **300 mm diameter**, with rotational, raster, and freeform raster polishing modes, which fits your 50–200 mm lens requirement.

Solaris Optics describes Q-flex 300 rotational mode as suitable for plano, spherical, and aspheric optics up to 300 mm diameter, with raster mode extending capability for more complex aspherics/freeforms.

Estimated price

Market level	Price range
China-made / local MRF machine	\$150,000–500,000
Used/refurbished MRF system	\$250,000–800,000
New premium QED Q-flex class	\$800,000–1,500,000+

MRF is usually the most expensive finishing machine in this chain. Annoying, but this is the “magic box” that saves you when asphere/freeform accuracy gets spicy.

D. Vacuum Multilayer AR / Hard Coating Machine

Used for AR coating, hard/durable coating, and multilayer dielectric coating.

Item	Recommended spec
Coating type	AR, hard/durable AR, multilayer dielectric
Technology	E-beam + IAD, PIAD, IBS, or sputtering
Chamber size	Enough for Ø 200 mm lenses and fixtures
Monitoring	Optical monitoring + quartz crystal monitor
Pre-clean	Ion/plasma source preferred
Uniformity	Ask for coating uniformity over Ø 200 mm
Automation	Recipe-controlled, data logging

Recommended methods

Technology	Use case
E-beam evaporation	Standard AR coating
IAD / PIAD	Durable AR coating, better density
Magnetron sputtering	Durable industrial optical coating
IBS	Premium laser/high-stability coating

Example suppliers

Supplier	Example system family
Bühler Leybold Optics	SYRUSpro, HELIOS, IBS systems
Satisloh	280-X / MC coating systems
OptoTech	Vacuum coating systems

Korea/China suppliers	E-beam/IAD/sputter optical coating machines
Denton Vacuum / Veeco / ULVAC	Thin-film coating platforms

Public listings show optical/lens vacuum coating machines around **\$70,000–100,000** for lower-cost systems, while more advanced modern AR coating equipment has historically been cited in the **\$500,000–1,000,000** range. Used AR/hard-coat equipment also appears in the market: one used Satisloh/Satis 280X listing showed **\$200,000**, and an Optimal UCS 150 listing showed **\$45,000**.

Estimated price

Market level	Price range
Used small AR/hard coating system	\$45,000–250,000
New China-made optical coating machine	\$70,000–250,000
New IAD/PIAD production coating system	\$300,000–1,000,000+
New IBS / premium sputtering coating system	\$1,000,000–3,000,000+

2. Required Support Equipment

You should not buy only the four main machines. For $R_a \leq 2$ nm and multilayer coating, metrology and cleaning are not optional.

E. Ultrasonic Cleaning Line

Item	Recommended spec
Tanks	4–8 tanks
Stages	Detergent, DI rinse, ultrasonic rinse, hot DI, drying
Drying	Filtered hot air / spin drying
Material	Stainless steel
Cleanliness	Coating-grade cleaning

Estimated price

Market level	Price range
Small manual ultrasonic cleaner	\$2,000–15,000
Semi-automatic optical cleaning line	\$30,000–120,000
Fully automatic cleaning line	\$120,000–400,000+

F. Interferometer for Surface Form

Needed for spherical, plano, aspheric with CGH/null optics, and transmitted wavefront inspection.

Item	Recommended spec
Type	Fizeau interferometer
Aperture	4 in / 6 in / 12 in depending optics
Use	Surface form, transmitted wavefront
Accessories	Transmission spheres, flats, CGH for aspheres

Example suppliers

Supplier	Example
Zygo	Verifire series
4D Technology	PhaseCam / AccuFiz
MÖLLER-WEDEL	Interferometers
Trioptics	Wavefront/metrology systems

Zygo Verifire systems are widely used for fast surface form and transmitted wavefront measurement of optical components.

Estimated price

Market level	Price range
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Used basic interferometer	\$30,000–150,000
New 4 in / 6 in production interferometer	\$150,000–400,000+
Large aperture / asphere-capable with CGH	\$400,000–1,000,000+

G. White-Light Interferometer / Optical Profiler

Needed for **$R_a \leq 2 \text{ nm}$** verification.

Item	Recommended spec
Type	White-light interferometer / coherence scanning interferometer
Vertical resolution	Sub-nm preferred
Measurement	R_a , R_q , surface texture, micro-roughness
Use	Before and after polishing/MRF

Example suppliers

Supplier	Example
Zygo	NewView series
Bruker	ContourGT / NPFlex
Keyence	3D optical profiler
Taylor Hobson	CCI series

Zygo's NewView 9000 is described as a coherence scanning interferometer with sub-nanometer precision, which is the type of tool you need for $R_a \leq 2 \text{ nm}$ validation.

Estimated price

Market level	Price range
Old/parts-only used profiler	\$2,000–15,000 , risky and usually incomplete

Good used optical profiler	\$40,000–100,000 ; one used NewView 7300 listing showed \$65,000
New production-grade optical profiler	\$100,000–250,000+

H. Spectrophotometer for Coating Inspection

Needed to verify AR coating, transmission, reflection, and multilayer performance.

Item	Recommended spec
Range	UV-Vis-NIR depending product
Example range	190–2500 nm or wider
Accessories	Reflectance accessory, integrating sphere
Use	Transmission, reflection, coating curve

Example suppliers

Supplier	Example
PerkinElmer	Lambda 850/950/1050
Agilent	Cary series
Shimadzu	UV-3600 series
Hitachi	U-series

A PerkinElmer Lambda 1050 is commonly used for high-performance UV/Vis/NIR optical measurements, including difficult samples such as optical coatings and thin-film filters.

Estimated price

Market level	Price range
Used UV-Vis-NIR spectrophotometer	\$20,000–80,000
Used Lambda 1050 example	Around \$48,500 in one used listing
New high-end UV-Vis-NIR system	\$80,000–200,000+

I. Centering / Edging / Chamfering Machine

Needed after polishing to control lens outer diameter, bevel, and optical/mechanical axis alignment.

Item	Recommended spec
Diameter	50–200 mm
Function	Centering, edging, bevel/chamfer
Lens type	Spherical/aspheric/plano
Precision	Optical centering capability

Estimated price

Market level	Price range
Basic lens edging machine	\$5,000–30,000
Precision CNC centering/edging machine	\$80,000–300,000
Premium automated system	\$300,000–600,000+

Do not confuse low-cost eyeglass edgers with precision optical centering machines. Eyeglass machines are cheap, but they are not enough for high-precision 50–200 mm optical glass production.

3. Budget by Production Level

Option 1 — Low-Cost Semi-Automatic Line

Good for learning, prototypes, low volume, and less demanding plano/spherical optics.

Equipment	Budget
CNC optical grinder	\$80,000–200,000
CNC optical polisher	\$50,000–150,000
MRF machine	\$150,000–400,000

Vacuum coating machine	\$70,000–250,000
Cleaning line	\$20,000–80,000
Interferometer	\$50,000–150,000
Optical profiler	\$40,000–100,000
Spectrophotometer	\$20,000–80,000
Centering/edging	\$30,000–120,000

Total rough budget: \$510,000–1,530,000

This is the cheapest realistic version. It may work, but the risk is inconsistent quality for aspheric/freeform and $Ra \leq 2$ nm.

Option 2 — Professional Semi-Automatic Precision Optics Line

Best balance for your specification.

Equipment	Budget
5-axis CNC grinder	\$300,000–700,000
CNC sub-aperture polisher	\$250,000–600,000
MRF correction machine	\$500,000–1,200,000
IAD/PIAD vacuum coating system	\$300,000–900,000
Cleaning line	\$80,000–200,000
Fizeau interferometer	\$150,000–400,000
White-light interferometer/profiler	\$100,000–250,000
Spectrophotometer	\$80,000–180,000
Centering/edging	\$120,000–300,000
Fixtures, tooling, coolant, slurry, installation	\$100,000–300,000

Total rough budget: \$1.98M–5.03M

This is the line I would recommend for your requirement.

Option 3 — Premium Full-Automatic / High-End Freeform Line

Best for serious production of aspheric/freeform optics, tight form accuracy, high yield, and coating repeatability.

Equipment	Budget
Premium CNC grinding cell	\$600,000–1,200,000
Premium CNC polishing cell	\$600,000–1,200,000
MRF Q-flex class system	\$900,000–1,500,000+
IBS / premium sputtering coating	\$1,000,000–3,000,000+
Robotic loading / automation	\$300,000–1,000,000
Automated cleaning	\$150,000–400,000
Advanced interferometry + CGH/freeform metrology	\$500,000–1,500,000
Optical profiler	\$150,000–300,000
Spectrophotometer	\$100,000–250,000
MES/process traceability	\$100,000–500,000

Total rough budget: \$4.4M–10.85M+

This is for high-end aerospace, semiconductor, laser, or premium imaging optics. Fancy, expensive, very capable.

4. My Recommended Procurement List

For your stated spec, I would ask vendors for this line:

Priority	Equipment	Recommended choice
1	CNC grinder	5-axis CNC optical grinder, Ø 5–200 mm or larger
2	CNC polisher	5-axis sub-aperture polisher, Ø 5–200 mm

3	MRF	MRF system supporting Ø 200 mm+, rotational + raster/freeform
4	Coating	IAD/PIAD vacuum coating system with optical monitoring
5	Cleaning	Semi/fully automatic ultrasonic cleaning line
6	Form metrology	Fizeau interferometer + asphere/freeform accessories
7	Roughness metrology	White-light interferometer / optical profiler
8	Coating metrology	UV-Vis-NIR spectrophotometer
9	Finishing	CNC centering/edging/chamfering machine

5. Price Summary Table

Equipment	Low-cost / used	Professional	Premium
CNC optical grinding	\$80k–250k	\$300k–800k	\$800k+
CNC optical polishing	\$50k–200k	\$250k–700k	\$700k+
MRF correction	\$150k–500k	\$500k–1.2M	\$1.2M+
Vacuum coating	\$70k–250k	\$300k–1M	\$1M–3M+
Cleaning line	\$20k–80k	\$80k–200k	\$200k–400k+
Interferometer	\$30k–150k	\$150k–400k	\$400k–1M+
Optical profiler	\$40k–100k	\$100k–250k	\$250k+
Spectrophotometer	\$20k–80k	\$80k–200k	\$200k+

Centering/edging	\$30k–120k	\$120k–300k	\$300k+
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Best Practical Budget

For your requirement, I would not plan below **\$2M** unless you accept used equipment, China-made systems, or lower automation.

My practical recommendation:

- 1 Professional semi-automatic line:
- 2 Budget: around \$2M–5M

Best process/equipment combination:

- 1 5-axis CNC optical grinder
- 2 + 5-axis CNC sub-aperture polisher
- 3 + MRF Ø 300 mm class machine
- 4 + IAD/PIAD vacuum AR coating system
- 5 + ultrasonic cleaning
- 6 + interferometer
- 7 + white-light profiler
- 8 + UV-Vis-NIR spectrophotometer

That setup gives you the best chance to repeatedly produce **50–200 mm optical glass spherical/aspheric/freeform/plano lenses with $R_a \leq 2$ nm and multilayer AR/hard coating.**

okay, give me the detailed datasheet for each equipment

I'll give this as vendor-style datasheets: purpose, suitable specs, machine requirements, process capability, utilities, inspection outputs, and supplier/model examples. I'll keep the numbers as realistic planning specs because exact machine datasheets vary by vendor and model.

Thought for 10s

Below is a **detailed procurement-style datasheet package** for each equipment category in the recommended line:

- 1 CNC optical grinding
- 2 → CNC optical polishing
- 3 → MRF correction
- 4 → vacuum multilayer coating
- 5 → cleaning
- 6 → metrology
- 7 → centering / edging

I'm using **representative production specs** suitable for your lens requirement, not locking you to one brand. Where I mention real examples, I cite the public source.

1. CNC Optical Grinding Machine Datasheet

Recommended machine type

5-axis CNC optical grinding machine

Representative example: **Satisloh SPM-200 class**. Satisloh lists the SPM-200 as an optical grinder for workpieces from **Ø 5–200 mm**, and its SPM/SPS-200 brochure describes support for spheres, aspheres, freeforms, edges, chamfers, and multiple spindle/tool configurations.

Application

Used for:

1	Rough curve generation
2	Fine grinding
3	Aspheric grinding
4	Freeform grinding
5	Edge/chamfer pre-processing
6	Plano surface preparation

Recommended technical specification

Item	Recommended requirement
Workpiece diameter	50–200 mm
Supported optics	Spherical, aspheric, freeform, plano
Material compatibility	BK7, fused silica, crown glass, flint glass, optical ceramics if needed
Machine axes	5-axis preferred
Work spindle	High-precision direct-drive spindle
Tool spindle	Water-cooled grinding spindle
Tooling	Diamond cup wheels, peripheral wheels, form wheels
Coolant	Temperature-controlled grinding coolant
Tool wear compensation	Required
CNC control	Recipe/program-based process control
Simulation	3D process simulation preferred

Automation	Manual loading minimum; robot loading optional
Enclosure	Fully enclosed wet grinding chamber

Process capability target

Output after grinding	Target
Geometry	Near-net optical shape
Surface condition	Uniform fine-ground surface
Subsurface damage	Minimized by fine grinding sequence
Polishing allowance	Controlled and repeatable
Edge chips	No chips beyond drawing tolerance
Form error	Within CNC polishing/MRF correction range

Grinding process flow

1	Blank loading
2	→ rough centering
3	→ rough grinding
4	→ semi-fine grinding
5	→ fine grinding
6	→ tool wear compensation
7	→ washing
8	→ form/thickness inspection

Key options to request from supplier

Option	Why important
Air-bearing workpiece spindle	Better surface finish and lower vibration
Multiple tool stations	Allows rough + fine tools in one setup
Freeform grinding software	Required for non-rotational lenses
Tool wear compensation	Important for stable production
Coolant temperature control	Prevents thermal error
Automation interface	Needed for semi/full automation

Utilities to ask vendor

Utility	Typical requirement to confirm
Power	3-phase industrial power
Compressed air	Clean, dry compressed air
Water/coolant	Grinding coolant system
Exhaust/mist control	Required for wet grinding
Foundation	Vibration-controlled floor preferred
Environment	Temperature-stable production room

Acceptance test

Ask the supplier to demonstrate:

- 1

2

3

4

5

6

7
1.

2.

3.

4.

5.

6.

7.
- Ø 50–200 mm optical glass processing

BK7 and fused silica grinding

Aspheric or freeform sample capability

Tool wear compensation result

Surface after fine grinding

Repeatability over multiple parts

Process report and machine log export

2. CNC Optical Polishing Machine Datasheet

Recommended machine type

5-axis CNC sub-aperture optical polishing machine

Representative example: **Satisloh SPS-200 class**. Satisloh describes the SPS-200 as a precision optics polisher for **spheres, aspheres, and freeform surfaces**, with a working range of **5–200 mm**.

Application

Used for:

- 1

2

3

4

5
- Removing fine grinding marks

Reducing surface roughness

Improving form accuracy

Preparing surface for MRF correction

Preparing surface for coating

Recommended technical specification

Item	Recommended requirement
Workpiece diameter	50–200 mm
Supported optics	Spherical, aspheric, freeform, plano
Polishing mode	Sub-aperture polishing preferred
Axes	5-axis preferred
Tooling	Pitch, polyurethane, bonnet, adaptive polishing tools
Slurry	Cerium oxide, colloidal silica, or material-specific slurry
Software	Dwell-time / tool-path control
Measurement input	Error-map import preferred
Automation	Manual or robotic loading
Process control	Recipe-based polishing process
Cleaning between passes	Required

Process capability target

Output after polishing	Recommended target
Surface roughness	Near $R_a \leq 2$ nm before MRF, depending material
Form error	Low enough for MRF final correction
Scratches/sleeks	None visible under inspection
Mid-spatial frequency	Controlled
Edge roll-off	Controlled
Surface readiness	Ready for MRF and coating after cleaning

Polishing process flow

- 1 | Fine-ground lens loading
- 2 | → pre-polishing

- 3 → cleaning
- 4 → surface measurement
- 5 → correction polishing pass
- 6 → second measurement
- 7 → final polish
- 8 → clean and inspect

Recommended polishing methods by lens type

Lens type	Recommended polishing method
Plano	Double-side polishing or CNC flat polishing
Spherical	CNC spherical polishing
Aspheric	CNC sub-aperture polishing
Freeform	5-axis CNC sub-aperture polishing

Key options to request

Option	Why important
Adaptive polishing tool	Useful for aspheres/freeforms
Dwell-time software	Needed for deterministic polishing
Error-map import	Required for closed-loop correction
Multiple tool sizes	Helps control local and global error
Temperature control	Improves repeatability
Process data logging	Needed for traceability

Acceptance test

Ask supplier to provide:

- 1 1. Surface roughness report after polishing
- 2 2. Form error before and after polishing
- 3 3. Sample aspheric/freeform correction result
- 4 4. Slurry/tool recommendation for BK7 and fused silica
- 5 5. Repeatability report
- 6 6. Defect inspection report

3. MRF Correction Machine Datasheet

Recommended machine type

Magnetorheological finishing machine, Ø 200 mm+ capacity.

Representative example: **QED Q-flex MRF-300 class**. QED describes Q-flex MRF-300 as supporting optics up to about **300 mm diameter**, with rotational, raster, and freeform raster polishing modes.

Application

Used for:

1	Final deterministic figure correction
2	Local high-spot removal
3	Aspheric correction
4	Freeform correction
5	Mid-spatial error reduction
6	Final surface preparation before coating

Recommended technical specification

Item	Recommended requirement
Workpiece diameter	50–200 mm, machine capacity ≥ 200 mm
Preferred machine capacity	Ø 300 mm class
Supported surfaces	Plano, spherical, aspheric, freeform
Correction mode	Rotational + raster/freeform mode
Software	Dwell-time correction from measured error map
Metrology input	Interferometer/profiler/CMM map import
Fluid	Magnetorheological polishing fluid
Spot control	Adjustable polishing spot size preferred
Automation	Semi-auto setup; automated recipe execution
Data output	Correction report and process history

MRF correction flow

1	Clean polished lens
2	→ measure surface error

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3 → generate error map
4 → calculate dwell-time map
5 → run MRF correction
6 → clean lens
7 → re-measure
8 → repeat if necessary

```

Process capability target

Output after MRF	Target
Surface figure	Corrected to final optical tolerance
Local form error	Reduced
Surface roughness	Compatible with $R_a \leq 2$ nm target
Mid-spatial frequency	Improved/controlled
Freeform error	Corrected using measured map
Coating readiness	Surface clean, stable, defect-free

Key options to request

Option	Why important
Freeform raster mode	Required for freeform lenses
Multiple spot sizes	Helps control different spatial frequencies
Error-map software	Required for deterministic correction
Metrology compatibility	Must work with your interferometer/profiler
Fluid recipe support	Different glass types may need different recipes
Edge correction tools	Important for Ø 50–200 mm lenses

Acceptance test

Ask supplier to show:

```

1 1. Before/after MRF correction map
2 2. Aspheric/freeform correction example
3 3. Surface roughness after MRF
4 4. Repeatability on same lens type
5 5. Edge-zone correction capability
6 6. Compatible metrology file formats

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Important purchasing warning

MRF should not be bought alone. It needs a strong metrology system. Bad measurement data creates bad MRF correction. The MRF machine is only as good as the measurement loop feeding it.

4. Vacuum Multilayer Coating Machine Datasheet

Recommended machine type

Vacuum optical coating machine with IAD / PIAD

Representative example: **Bühler Leybold Optics SYRUSpro class**. Bühler describes the SYRUSpro series as available in chamber sizes from **710 mm to 2800 mm**, and its technical data lists **ion-assisted deposition and plasma-ion-assisted deposition** with common coating materials including dielectrics, metals, fluorides, and sulfides.

Application

Used for:

- 1 AR coating
- 2 Hard/durable AR coating
- 3 Multilayer dielectric coating
- 4 Protective coating
- 5 UV/VIS/NIR optical coating

Recommended technical specification

Item	Recommended requirement
Substrate size	Must support Ø 200 mm lenses
Coating method	E-beam + IAD/PIAD preferred
Premium option	IBS or sputtering for highest stability
Chamber size	≥ 700 mm inner chamber recommended
Deposition sources	Multiple material sources preferred
Ion source	Required for dense/durable coating
Monitoring	Optical monitoring + quartz crystal monitoring
Rotation	Planetary rotation preferred for uniformity

Pre-cleaning	Ion/plasma cleaning
Coating recipe	Programmable multilayer recipe
Data logging	Required for traceability

Coating process flow

1	Final inspection before coating
2	→ ultrasonic cleaning
3	→ DI rinse and drying
4	→ load coating fixture
5	→ vacuum pump down
6	→ ion/plasma cleaning
7	→ deposit multilayer stack
8	→ optical monitoring
9	→ cooling
10	→ unload
11	→ spectrophotometer inspection

Coating capability target

Output	Target
Coating type	AR / hard AR / multilayer dielectric
Wavelength	Customer-defined, for example VIS/NIR/UV
Reflection	Customer-defined, e.g. $R_{avg} < 0.5\%$ per surface
Uniformity	Must be specified over $\varnothing 200$ mm
Durability	Adhesion, humidity, abrasion test
Defects	Low coating pits, stains, particles
Stress	Controlled to avoid wavefront distortion

Coating materials

Layer type	Common materials
Low-index	SiO_2 , MgF_2
High-index	Ta_2O_5 , TiO_2 , Nb_2O_5 , HfO_2
Durable/protective	SiO_2 , Al_2O_3 , dense oxide layers

Optional top layer	Hydrophobic / oleophobic layer
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Key options to request

Option	Why important
IAD or PIAD source	Higher coating density and durability
Optical monitoring	Better spectral accuracy
Planetary substrate holder	Better coating uniformity
Ion pre-cleaning	Better adhesion
Multiple source pockets	More complex multilayer stacks
Clean loading interface	Reduces coating defects
Recipe data export	Quality traceability

Acceptance test

Ask supplier to provide:

- 1 1. AR coating curve report
- 2 2. Coating uniformity map over Ø 200 mm
- 3 3. Adhesion test result
- 4 4. Humidity/temperature cycling result
- 5 5. Abrasion test result
- 6 6. Before/after wavefront distortion report
- 7 7. Coating defect inspection report

5. Ultrasonic Cleaning Line Datasheet

Recommended machine type

Multi-stage ultrasonic optical cleaning line

Representative example: ESMA describes the E700 as a PLC-controlled automatic ultrasonic system for **cleaning, rinsing, and drying optical lenses**, intended to help produce residue-free lenses ready for coating.

Application

Used before:

- 1 MRF correction

- 2 Final inspection
- 3 Vacuum coating
- 4 Packaging

Recommended technical specification

Item	Recommended requirement
Stages	4–8 tanks
Cleaning method	Ultrasonic + DI rinse
Control	PLC-controlled
Tank material	Stainless steel
Water	DI water compatible
Filtration	Recirculation filtration preferred
Drying	Hot filtered air or vacuum drying
Lens handling	Basket or custom fixture
Cleanliness	Coating-ready surface

Recommended cleaning sequence

- 1 Alkaline detergent ultrasonic clean
- 2 → DI water ultrasonic rinse
- 3 → DI overflow rinse
- 4 → final DI rinse
- 5 → optional IPA/solvent rinse
- 6 → filtered hot-air dry
- 7 → clean inspection

Key options to request

Option	Why important
Multiple ultrasonic frequencies	Different contamination removal
DI water recirculation	Stable cleanliness
Particle filtration	Reduces coating defects
Automatic basket transfer	Lower operator contamination
HEPA drying	Coating-grade cleanliness

Conductivity monitoring	DI water quality control
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Acceptance test

Ask supplier to demonstrate:

- 1 1. No visible residue after cleaning
- 2 2. Coating adhesion after cleaning
- 3 3. Particle count target
- 4 4. Water-break-free surface
- 5 5. Compatibility with BK7/fused silica/coated optics

6. Fizeau Interferometer Datasheet

Recommended machine type

Laser Fizeau interferometer

Representative example: ZYGO Verifire HD/HDX class. ZYGO says Verifire systems are used for surface form and transmitted wavefront measurement, and Verifire HD/HDX systems are aimed at high-performance optical components including mid-spatial frequency characterization.

Application

Used for:

- 1 Surface form measurement
- 2 Transmitted wavefront measurement
- 3 Plano surface flatness
- 4 Spherical surface form
- 5 Asphere measurement with CGH/null optics
- 6 Process feedback to polishing/MRF

Recommended technical specification

Item	Recommended requirement
Aperture	4 in minimum; 6 in preferred for Ø 200 mm with setup strategy
Laser wavelength	633 nm HeNe typical
Measurement type	Surface form and transmitted wavefront
Accessories	Transmission flats, transmission spheres, mounts

Asphere support	CGH/null optics if needed
Software	Zernike, PV, RMS, slope, map export
Export format	Compatible with MRF correction software
Environment	Vibration-isolated optical table

Key options to request

Option	Why important
6-inch aperture	More useful for large optics
Transmission flats/spheres	Needed for different lens types
CGH support	Needed for aspheres/freeforms
Vibration isolation	Stable measurement
Stitching capability	Useful for larger/complex optics
Data export	Needed for MRF correction loop

Acceptance test

Ask for:

- 1 1. Calibration certificate
- 2 2. Reference flat measurement
- 3 3. Repeatability test
- 4 4. Measurement map export
- 5 5. Compatibility with MRF software
- 6 6. Sample lens measurement report

7. White-Light Interferometer / Optical Profiler Datasheet

Recommended machine type

Coherence scanning interferometer / white-light optical profiler

Representative example: ZYGO NewView 9000 class. ZYGO describes the NewView 9000 as a 3D optical profiler using coherence scanning interferometry with sub-nanometer precision, suitable for surface topography measurement.

Application

Used for:

1	Ra measurement
2	Rq measurement
3	Micro-roughness
4	Surface texture
5	Tool marks
6	Polishing defects
7	MRF surface validation

Recommended technical specification

Item	Recommended requirement
Vertical resolution	Sub-nanometer preferred
Measurement mode	White-light / coherence scanning interferometry
Roughness parameters	Ra, Rq, Sa, Sq
Objectives	Multiple magnifications
Stage	Motorized XY preferred
Software	ISO roughness analysis
Data output	Surface maps and roughness reports
Environment	Vibration-isolated bench

Measurement requirement for your product

For your requirement:

1	Surface roughness: $Ra \leq 2 \text{ nm}$
---	---

You should require:

1	1. Measurement before coating
2	2. Measurement after final polish/MRF
3	3. Defined scan area
4	4. Defined filtering method
5	5. Defined roughness standard
6	6. Calibration record

Key options to request

Option	Why important
--------	---------------

Sub-nm vertical precision	Needed for $R_a \leq 2$ nm
Multiple objectives	Different spatial scales
Motorized stitching	Larger area roughness maps
ISO 25178 support	3D surface texture standard
Vibration isolation	Stable roughness measurement

Acceptance test

Ask supplier to provide:

1	1. Calibration artifact measurement
2	2. R_a repeatability test below 2 nm
3	3. Sample polished glass measurement
4	4. Measurement recipe export
5	5. Operator training package

8. UV-VIS-NIR Spectrophotometer Datasheet

Recommended machine type

High-performance UV/VIS/NIR spectrophotometer

Representative example: **PerkinElmer Lambda 1050 / 1050+ class**. PerkinElmer describes the LAMBDA 1050+ as a high-performance UV/Vis/NIR system for a wide range of sample types, and Lambda 1050 documentation states measurement capability up to **3300 nm** with triple-detector sensitivity in the NIR.

Application

Used for:

1	AR coating verification
2	Transmission measurement
3	Reflection measurement
4	Multilayer coating curve
5	Filter performance
6	Coating batch QC

Recommended technical specification

Item	Recommended requirement
Wavelength range	175–3300 nm preferred

Optical configuration	Double-beam preferred
Detector	UV/VIS/NIR detector set
Accessories	Reflectance accessory, integrating sphere
Measurement	T%, R%, absorbance
Sample fixture	Lens holder for Ø 50–200 mm
Software	Coating curve export
Data output	CSV/PDF report

Key options to request

Option	Why important
Specular reflectance accessory	Needed for AR coating R%
Integrating sphere	Useful for scattering/diffuse measurement
Large sample compartment	Needed for Ø 200 mm lenses
Variable angle accessory	Needed if AOI is not 0°
Polarizer accessory	Needed for S/P coating analysis

Acceptance test

Ask supplier to show:

1	1. Transmission curve for coated sample
2	2. Reflection curve for coated sample
3	3. Measurement repeatability
4	4. Baseline stability
5	5. Exportable QC report

9. CNC Centering / Edging / Chamfering Machine Datasheet

Recommended machine type

Precision CNC optical centering and edging machine

Application

Used for:

- | | |
|---|---|
| 1 | Final outside diameter control |
| 2 | Optical axis to mechanical axis alignment |
| 3 | Edge bevel/chamfer |
| 4 | Chip prevention |
| 5 | Mounting preparation |

Recommended technical specification

Item	Recommended requirement
Workpiece diameter	50–200 mm
Supported optics	Plano, spherical, aspheric lenses
Operation	Centering, edging, beveling, chamfering
Tooling	Diamond edging wheels
Coolant	Wet grinding coolant
Axis control	CNC controlled
Centering method	Optical/mechanical centering system
Automation	Manual loading or robot-ready

Process capability target

Output	Target
Final OD	According to drawing
Edge quality	No chips beyond tolerance
Chamfer	Uniform bevel
Centering	Optical axis aligned to mechanical OD
Coating readiness	Edge clean and stable

Key options to request

Option	Why important
--------	---------------

Optical centering measurement	Required for precision lenses
Chamfer control	Prevents coating/handling chips
Recipe control	Repeatability
Soft contact fixture	Avoids polished surface damage
Coolant filtration	Prevents scratches

Acceptance test

Ask supplier for:

- 1 1. Centering accuracy report
- 2 2. OD repeatability report
- 3 3. Edge-chip inspection
- 4 4. Chamfer consistency
- 5 5. Demonstration on Ø 200 mm glass lens

10. Recommended “Complete Line” Datasheet Summary

Process step	Equipment	Minimum recommended class
Shape generation	5-axis CNC optical grinder	Ø 50–200 mm, spherical/aspheric/freeform
Surface finishing	5-axis CNC optical polisher	Sub-aperture, error-map capable
Final correction	MRF correction machine	Ø 200 mm+, rotational + raster/freeform
Coating	Vacuum multilayer coater	IAD/PIAD, optical monitoring
Cleaning	Ultrasonic cleaning line	4–8 tank, DI rinse, filtered dry
Form metrology	Fizeau interferometer	4–6 inch, CGH/null support
Roughness metrology	White-light optical profiler	Sub-nm vertical precision
Coating metrology	UV/VIS/NIR spectrophotometer	175–3300 nm preferred
Mechanical finishing	CNC centering/edging	Ø 50–200 mm, optical centering

Recommended Vendor RFQ Text

You can send this to vendors:

```
1 We are looking for equipment for precision optical glass lens
2 production.
3 Target product:
4 - Lens diameter: 50-200 mm
5 - Lens types: spherical, aspheric, freeform, plano
6 - Materials: BK7, fused silica, crown/flint optical glass
7 - Surface roughness target:  $R_a \leq 2$  nm before coating
8 - Production mode: semi-automatic or full automatic
9 - Coating: AR, hard/durable AR, multilayer dielectric coating
10
11 Required process:
12 CNC grinding → CNC polishing → MRF correction → ultrasonic cleaning →
13 vacuum multilayer coating → final metrology.
14 Please provide detailed datasheet, machine configuration, process
15 capability, utility requirements, automation options, metrology
16 integration, compatible materials, sample inspection reports,
17 installation requirements, training package, warranty, spare parts
18 list, delivery time, and quotation.
19
20 Please also confirm whether your equipment can support:
21 1. Ø 50-200 mm optical glass lenses
22 2. Spherical/aspheric/freeform/plano geometry
23 3. BK7 and fused silica processing
24 4.  $R_a \leq 2$  nm surface roughness process capability
25 5. Error-map feedback from interferometer/profiler to polishing/MRF
26 6. Multilayer AR/hard coating for customer-defined wavelength ranges
27 7. Process data logging and production traceability
```

My Recommended Baseline Configuration

For your case, the best practical configuration is:

```
1 1. 5-axis CNC optical grinder, Ø 5-200 mm or larger
2 2. 5-axis CNC sub-aperture polisher, Ø 5-200 mm or larger
3 3. MRF machine, Ø 300 mm class
4 4. IAD/PIAD vacuum multilayer coating system, chamber ≥ 700 mm
5 5. Automatic ultrasonic optical cleaning line
6 6. 6-inch Fizeau interferometer with CGH/null optics option
7 7. White-light optical profiler with sub-nm vertical precision
8 8. UV/VIS/NIR spectrophotometer with reflectance accessory
9 9. Precision CNC centering/edging/chamfering machine
```

This is the configuration I'd put into a serious RFQ for **50–200 mm optical glass lenses with $R_a \leq 2$ nm and multilayer AR/hard coating.**

okay, please

Thought for a couple of seconds

Sure — here is a **professional RFQ / inquiry document** you can send to equipment vendors.

RFQ: Precision Optical Glass Lens Production Equipment Line

1. Project Overview

We are looking for equipment for a precision optical glass lens production line.

The target production process is:

1	CNC optical grinding
2	→ CNC optical polishing
3	→ MRF correction
4	→ ultrasonic cleaning
5	→ vacuum multilayer coating
6	→ final optical inspection

The production line should support semi-automatic or full-automatic operation.

2. Target Product Specification

Item	Requirement
Lens diameter	50–200 mm
Lens types	Spherical, aspheric, freeform, plano
Materials	BK7, fused silica, crown glass, flint glass, other optical glass
Surface roughness	$Ra \leq 2$ nm before coating
Coating	AR coating, hard/durable coating, multilayer dielectric coating
Production mode	Semi-automatic or full-automatic
Application	Precision optics / imaging optics / industrial optics / laser optics, if applicable

3. Required Equipment Scope

Please provide quotation and datasheet for the following equipment.

3.1 CNC Optical Grinding Machine

Required capability:

Item	Requirement
Workpiece diameter	50–200 mm
Supported surfaces	Spherical, aspheric, freeform, plano
Materials	BK7, fused silica, crown/flint optical glass
Axis configuration	5-axis preferred
Process	Rough grinding, fine grinding, curve generation
Tooling	Diamond grinding wheels
Coolant	Temperature-controlled coolant system
Automation	Manual, semi-auto, or robot loading option
Data logging	Required

Please confirm:

1	1. Maximum and minimum workpiece diameter
2	2. Supported lens geometries
3	3. Achievable form accuracy after grinding
4	4. Recommended grinding wheel types
5	5. Tool wear compensation function
6	6. Processing examples for BK7 and fused silica
7	7. Utility requirements
8	8. Installation and training requirements

3.2 CNC Optical Polishing Machine

Required capability:

Item	Requirement
Workpiece diameter	50–200 mm
Supported surfaces	Spherical, aspheric, freeform, plano
Polishing method	CNC sub-aperture polishing preferred
Tooling	Pitch, polyurethane, bonnet, or equivalent polishing tools

Slurry	Cerium oxide / colloidal silica / material-specific slurry
Software	Dwell-time correction / tool-path control
Metrology integration	Error-map import preferred
Automation	Semi-auto or full-auto option

Please confirm:

- 1 1. Achievable surface roughness after polishing
- 2 2. Achievable form accuracy after polishing
- 3 3. Compatibility with aspheric and freeform optics
- 4 4. Compatible metrology data formats
- 5 5. Recommended slurry and polishing tools
- 6 6. Process repeatability
- 7 7. Sample inspection report

3.3 MRF Correction Machine

Required capability:

Item	Requirement
Workpiece diameter	50–200 mm
Preferred capacity	Ø 300 mm class
Supported surfaces	Plano, spherical, aspheric, freeform
Correction mode	Rotational + raster/freeform mode
Software	Dwell-time map generation
Metrology integration	Interferometer/profiler/CMM error-map import
Output	Final deterministic figure correction
Process record	Required

Please confirm:

- 1 1. Maximum workpiece diameter
- 2 2. Supported optical geometries
- 3 3. Achievable figure correction accuracy
- 4 4. Achievable surface roughness after MRF
- 5 5. Supported error-map file formats
- 6 6. Edge correction capability
- 7 7. Example before/after MRF correction report

3.4 Vacuum Multilayer Coating Machine

Required capability:

Item	Requirement
Coating type	AR, hard/durable AR, multilayer dielectric
Technology	E-beam + IAD/PIAD preferred
Premium option	IBS / sputtering optional
Substrate size	Must support Ø 200 mm lenses
Monitoring	Optical monitoring + quartz crystal monitoring
Pre-clean	Ion/plasma cleaning preferred
Rotation	Planetary rotation preferred
Data logging	Required
Recipe control	Required

Please confirm:

1. Coating technology: evaporation, IAD, PIAD, IBS, or sputtering
2. Chamber size
3. Maximum lens diameter
4. Coating uniformity over Ø 200 mm
5. Supported coating materials
6. AR coating wavelength range capability
7. Hard/durable coating capability
8. Adhesion, humidity, abrasion test data
9. Coating curve sample report
10. Maintenance and consumable requirements

3.5 Ultrasonic Optical Cleaning Line

Required capability:

Item	Requirement
Cleaning stages	4–8 tanks preferred
Cleaning type	Ultrasonic cleaning + DI rinse

Drying	Filtered hot air / HEPA drying preferred
Water	DI water compatible
Automation	Semi-auto or fully automatic
Application	Coating-grade optical cleaning

Please confirm:

1	1. Number of tanks
2	2. Cleaning sequence
3	3. Ultrasonic frequency
4	4. DI water requirement
5	5. Drying method
6	6. Lens fixture design
7	7. Cleanliness level
8	8. Compatibility with BK7 and fused silica

3.6 Fizeau Interferometer

Required capability:

Item	Requirement
Measurement	Surface form / transmitted wavefront
Aperture	4-inch minimum, 6-inch preferred
Lens support	Plano, spherical, aspheric with CGH/null option
Software	PV, RMS, Zernike, slope, map export
Export	Compatible with polishing/MRF correction software

Please confirm:

1	1. Aperture size
2	2. Measurement accuracy
3	3. Repeatability
4	4. Available transmission flats/spheres
5	5. CGH/null optics support
6	6. Data export formats
7	7. Calibration certificate
8	8. Recommended optical table and environment

3.7 White-Light Interferometer / Optical Profiler

Required capability:

Item	Requirement
Measurement	Ra, Rq, Sa, Sq, surface texture
Target roughness	Must verify $Ra \leq 2 \text{ nm}$
Vertical resolution	Sub-nanometer preferred
Stage	Motorized XY preferred
Software	ISO roughness analysis
Output	Roughness report and surface map

Please confirm:

1	1. Vertical resolution
2	2. Roughness measurement repeatability
3	3. Minimum measurable roughness level
4	4. Scan area
5	5. Objective lens options
6	6. Filtering method
7	7. Calibration standard
8	8. Sample report for polished optical glass

3.8 UV-VIS-NIR Spectrophotometer

Required capability:

Item	Requirement
Measurement	Transmission and reflection
Wavelength range	UV/VIS/NIR, preferably up to 2500–3300 nm
Accessories	Reflectance accessory required
Optional	Integrating sphere, variable angle accessory, polarizer
Sample size	Ø 50–200 mm lens support
Output	Coating curve report

Please confirm:

- 1 1. Wavelength range
- 2 2. Transmission measurement capability
- 3 3. Reflection measurement capability
- 4 4. Large lens fixture availability
- 5 5. Variable angle measurement option
- 6 6. Polarization measurement option
- 7 7. Software report format
- 8 8. Calibration requirement

3.9 CNC Centering / Edging / Chamfering Machine

Required capability:

Item	Requirement
Workpiece diameter	50–200 mm
Process	Centering, edging, beveling/chamfering
Material	Optical glass
Tooling	Diamond edging wheels
Coolant	Wet grinding coolant
Centering	Optical/mechanical centering support
Automation	Manual or semi-auto

Please confirm:

- 1 1. Maximum and minimum lens diameter
- 2 2. Centering accuracy
- 3 3. OD accuracy
- 4 4. Chamfer capability
- 5 5. Edge-chip control
- 6 6. Fixture type
- 7 7. Sample edging report

4. Required Vendor Submission

Please include the following in your quotation package:

- 1 1. Complete technical datasheet for each machine
- 2 2. Machine configuration and included options
- 3 3. Process capability statement
- 4 4. Supported materials and lens geometries
- 5 5. Sample production reports
- 6 6. Sample inspection reports

- 7 7. Utility requirements
- 8 8. Machine layout drawing
- 9 9. Automation options
- 10 10. Software functions and data export format
- 11 11. Consumables and spare parts list
- 12 12. Installation and commissioning plan
- 13 13. Operator training plan
- 14 14. Warranty terms
- 15 15. Maintenance schedule
- 16 16. Delivery time
- 17 17. Price quotation
- 18 18. Payment terms
- 19 19. After-sales service location
- 20 20. Reference customers or case studies, if available

5. Required Process Capability Guarantee

Please clearly state whether your proposed equipment line can support the following:

Requirement	Vendor confirmation
Ø 50–200 mm optical glass lenses	Yes / No
BK7 processing	Yes / No
Fused silica processing	Yes / No
Crown/flint glass processing	Yes / No
Spherical lens production	Yes / No
Aspheric lens production	Yes / No
Freeform lens production	Yes / No
Plano lens production	Yes / No
$R_a \leq 2$ nm before coating	Yes / No
MRF correction	Yes / No
AR multilayer coating	Yes / No
Hard/durable coating	Yes / No
Coating inspection	Yes / No
Process traceability	Yes / No
Semi-automatic operation	Yes / No

Full-automatic upgrade option	Yes / No
-------------------------------	----------

6. Requested Sample Demonstration

Before final purchase, we request a sample production or demonstration using the following sample lens:

```

1 | Material: BK7 or fused silica
2 | Diameter: 100 mm minimum, preferred 150-200 mm
3 | Lens type: spherical or aspheric
4 | Surface roughness target: Ra ≤ 2 nm before coating
5 | Coating: broadband AR coating
6 | Inspection: form report, roughness report, coating
  | transmission/reflection report

```

Please provide:

```

1 | 1. Grinding report
2 | 2. Polishing report
3 | 3. MRF correction before/after report
4 | 4. Surface roughness report
5 | 5. Coating curve report
6 | 6. Final inspection report
7 | 7. Defect inspection report

```

7. Recommended Budget Format

Please quote the machines separately and as a complete production line.

Equipment	Unit price	Options	Installation	Training	Delivery
CNC optical grinder					
CNC optical polisher					
MRF correction machine					
Vacuum coating machine					

Ultrasonic cleaning line					
Fizeau interferometer					
White-light profiler					
Spectrophoto meter					
Centering/edg ing machine					
Tooling/fixture s/consumable s					
Total package					

8. My Suggested Short Vendor Message

You can send this first:

Email

Subject

Inquiry for Precision Optical Glass Lens Production Equipment Line

Dear [Vendor Name],

We are looking for equipment for a precision optical glass lens production line.

Our target product specification is:

- Lens diameter: 50–200 mm
- Lens types: spherical, aspheric, freeform, and plano
- Materials: BK7, fused silica, crown/flint optical glass, and other optical glass
- Surface roughness target: $R_a \leq 2$ nm before coating

- Coating: AR, hard/durable AR, and multilayer dielectric coating
- Automation: semi-automatic or full-automatic production

The required process is:

CNC optical grinding → CNC optical polishing → MRF correction → ultrasonic cleaning → vacuum multilayer coating → final inspection.

Please provide your recommended equipment configuration, technical datasheets, process capability, quotation, delivery time, installation requirements, training package, warranty, and after-sales support information.

Please also confirm whether your equipment can support Ø 50–200 mm optical glass lenses, aspheric/freeform processing, $R_a \leq 2$ nm surface roughness, MRF correction, and multilayer AR/hard coating.

Best regards,
[Your Name]

For your project, this RFQ is strong enough to send to **Satisloh, OptoTech, QED, Bühler Leybold Optics, OptiPro, Zeeko, Chinese optical machine suppliers, and coating machine suppliers.**